

UBT839: DRUG DESIGN AND DEVELOPMENT

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective of this course is to provide comprehensive knowledge of the basics of the drug discovery and development in the area of pharmacognosy and natural products; traditional and complementary medicine, synthetic medicinal chemistry and development of modern and innovative therapeutic substances.

Syllabus

Introduction: Definition of drugs, Overview of drug discovery process, Economics of drug discovery process, Trends in drug discovery process.

Rationale of Drug Discovery: Medical needs, Target identification, Target validation, Receptors and assay development.

Bio-resources for Small Molecule Discovery: Bioprospecting, Plant natural products, Microbial secondary metabolites, Marine natural products.

Screening Strategies for Drug Leads: Bioassay guided isolation, High throughput assays for antimicrobial, anticancer, anti-diabetic and anti-hypercholesterolemia, combinatorial chemogenomics, combinatorial chemistry. Characterization of drug molecules using integrated technology (TLC, HPLC, MS, IR, NMR).

Complementary and Alternative Medicine: Ayurveda and Herbal Drugs, Definition, Trade scenario, Pharmacopoeial status of herbal drugs.

Biosimilars: Introduction to biologics, defining biosimilars, differences between biosimilars and generics, selected examples of approved biosimilars, technical challenges associated with production of biosimilar molecules, regulatory aspects of biosimilar molecules. Current status of biosimilars in different countries (Europe, USA).

Drug Development and Pre-Clinical Studies: Introduction to structure–activity relationships (SAR), Drug receptor interactions; enzyme inhibition and inactivation, *In vitro* and *in vivo* pharmacodynamic models, Therapeutic index, Pharmacokinetics - Microbial and animal models, Lipinski's rule, *In vitro* and *in silico* toxicological models, Drug formulations.

Drug Regulatory Operations and Drug Manufacturing: Role of Regulatory Authorities, US FDA, Regulatory applications viz. investigational new drug (IND), new drug application (NDA), Abbreviated New Drug Application (ANDA).

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Demonstrate current approaches and steps of global drug discovery, their advantages and limitations.
2. Demonstrate awareness of different disciplines which play an integral role in drug discovery and development process.
3. Comprehend the key role played by natural products and pharmacognosy in shaping the pharmaceutical industry
4. Develop understanding of drug targets, their role in drug discovery process and their interaction with natural and synthetic ligands.
5. Demonstrate the importance of quality control and regulatory aspects of drug development processes and good manufacturing of medicines

Text Books

1. Benjamin B Basic Principles of Drug Discovery and Development, Academic Press, 1st Edition, ISBN : 9780124115088 (2015).
2. Larsen PK, Leljifore T and Medsan U, Text books of Drug Design and Discovery, CRC Press (2009) 4th ed.
3. Hillisch A and Hilgenfeld R, Modern Methods of Drug Discovery, Birkhauser (2003).

Reference Books

1. Patwardhan B, Drug Discovery and Development - Traditional Medicine and Ethnopharmacology, New India Publishing (2007).
2. Rick NG, Drugs from Discovery to Approval, 2nd Edition, Wiley- Blackwell (2009).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.